EFFECT OF I/O OXIDE PROCESS OPTIMIZATION ON THE NBTI DEPENDENCE OF T_{inv} SCALING FOR A 20 NM BULK PLANAR REPLACEMENT GATE PROCESS

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***Abstract*—We present results on the beneficial effect of an additional thermal treatment on the NBTI aging of an I/O thick oxide process in 20nm Replacement Metal Gate (RMG) High-κ Metal Gate (HKMG) technology. It is shown that for an as-grown thermal thick oxide gate process, the NBTI induced device threshold voltage shift (ΔV_{th}) scales as T_{inv}^{-2} when stressed at a given gate voltage V_g. On the other hand, the ΔV_{th} dependence on V_g is $\Delta V_{th} \sim (V_g)^4$ for a given T_{inv}. These findings seem to be apparently inconsistent with well known power law NBTI dependence on E_{ox} ($\sim (V_g/T_{inv})$). The expected NBTI dependence on E_{ox} ($\Delta V_{th} \sim (V_g/T_{inv})^4$) is recovered when the as-grown thermal thick oxide is treated with an additional high temperature anneal. We have evaluated the NBTI induced ΔV_{th} time evolution, its recovery behavior, temperature and voltage dependence under DC and AC bias stress conditions with and without the oxide thermal treatment. Our observations over the two gate stack processes (as-grown I/O oxide with and without anneal treatment) support the NBTI physical picture of two uncorrelated contributions to the NBTI damage. Namely, shallow hole trap activation in process induced pre-existing (before stress) traps as well as deep hole traps and/or interface states generation. The last component is quasi permanent and is dominant in both gate stack processes. In particular, for the as-grown I/O oxide the NBTI damage is mainly due to an increase contribution of the quasi permanent component with increasing T_{inv}. This unexpected T_{inv} dependence is reduced by the additional anneal treatment. The implication of these findings on the NBTI characterization and modeling will be discussed.**

Keywords- Bias temperature instability (BTI), thick oxide process, thermal anneal, NBTI dependence on Eox, High-κ Replacement Metal Gate Transistor (RMG), technology qualification

I. INTRODUCTION

NBTI is a significant degradation mechanism for p-MOSFETs, which is dominated by interface quality, and associated with interface state generation and charge trap activation and/or generation at the IL/Silicon interface [1-2]. It is well known that, among the various MOSFET reliability wear-out mechanisms, NBTI is a limiting factor for p-MOSFETs scaling of High-k Metal Gate (HKMG) technologies. There are two aspects of the CMOS scaling contributing to this trend. One is the strong NBTI sensitivity to HKMG CMOS process optimization [3-5]. In particular the device IL/Silicon interface quality and process treatment strongly impact the relative contribution to the NBTI damage of two independent and uncorrelated components: a slowly recoverable (typically attributable to interface state (ΔN_{it}) and/or deep traps (ΔN_{dht}) generation) and a fast recoverable (typically given by pre existing shallow hole trap activation (ΔN_{sht})) part. Another well know NBTI feature is its dependence on the Gate oxide electric field (E_{ox}) and, hence, it increases with CMOS T_{inv} reduction [6], where T_{inv} is the gate stack thickness in inversion mode. For $E_{ox} \sim (V_g/T_{inv})$ we assume that the threshold voltage shift (ΔV_{th}) is proportional to $(V_g/T_{inv})^m$, where Vg is the voltage applied on the device gate and m is the power law exponent. In this case it is expected that the V_{th} shift dependence on T_{inv} is in the form of T_{inv}^{-m} for a particular V_g at stress.

In this paper, we will show, for the first time, how the thermal treatment of a 20nm I/O thick oxide process has a significant beneficial effect on the NBTI dependence on T_{inv} scaling. It is observed that, when stressed at a given V_g, the NBTI induced device threshold voltage shift (ΔV_{th}) of an as-grown I/O thick HKMG gate oxide follows an anomalous trend with T_{inv}. After the high temperature anneal, ΔV_{th} follows the expected trend from the NBTI dependence on E_{ox}, and this effect will be further discussed. Comparison of the V_{th} shift, its recovery behavior, the stress voltage dependence, and gate oxide T_{inv} dependence, under DC and AC bias conditions, between the oxide processes with and without the thermal anneal, will also be discussed to elucidate the observed benefit to the NBTI damage.

II. EXPERIMENTAL

The n & p I/O MOSFETs tested in this study were fabricated using a 20nm RMG process [4] and consisted of a thermally grown SiO_2 interfacial layer/HfO_2 dual dielectric stack, capped with appropriate workfunction electrodes for

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pMOSFETs. The high temperature thermal anneal was added after the growth of the I/O oxide process. Different Tinv thickness was obtained by varying the thermally grown SiO_2 interfacial layer. The BTI tests were conducted at different voltages and temperatures, under DC and AC bias. The device threshold voltage V_{th} change is carefully measured by a method known as fast single I_D spot measurement to minimize the recovery during measurement. Drain current I_D was recorded at test voltage during pre-determined stress intervals, and then was converted to threshold voltage V_{th} shift using the sub-threshold slope extracted from initial I_D-V_G curve. At each readout tests, after the initial I_D measurement from <1 ms to 2 ms, the device was kept at test mode up to 1000 s to recover, and BTI V_{th} shift recovery can be obtained for each readout tests. AC stress was done at 50% duty cycle and 100 kHz.

III. Results and Discussion

A. Comparison of VAE and Tinv dependence of NBTI under DC bias stress

Fig. 1 shows the voltage dependence of the DC NBTI V_{th} shift on the I/O pMOSFETs with the same T_{inv} thickness but two different gate oxide processes, respectively, the as-grown thermal gate oxide with and without an additional high temperature anneal. The NBTI damage follows a power law dependence on V_g with similar voltage acceleration exponent, i.e., VAE ~ 4. This V_g dependence is consistent with previously reported values.

However, when pMOSFETs with various T_{inv} thicknesses were stressed at the same gate bias voltage, a different dependence of the ΔV_{th} on T_{inv} is observed. As shown in Fig. 2, the device with as-grown thermal gate oxide shows a much weaker T_{inv} dependence as $\Delta V_{th} \sim T_{inv}^{-2}$, instead of the expected T_{inv}^{-4} dependence observed after the extra thermal anneal. Obviously, these findings undermine the E_{ox} dependence of the NBTI degradation. These results seem to suggest the NBTI damage is dominated by a defect component whose contribution increases with T_{inv} for the devices with as-grown thermal gate oxide process. Once the gate oxide is treated with high temperature thermal anneal, the DC NBTI V_{th} shift follows the expected field dependence, as shown in Fig. 2, and the NBTI V_{th} shift is also reduced significantly, by ~2x as given in Fig. 3.

The similar NBTI time evolution (~0.24) of the devices with and without the extra high temperature anneal excludes the possibility of the recoverable fast hole trapping as a contributing factor to the weak T_{inv} dependence and suggesting a key role of the permanent component (ΔN_{it} and/or deep bulk trap ΔN_{dot}) to these dependencies. This hypothesis is also consistent with the similar NBTI activation energy (E_a~0.15eV) measured from devices with and without gate oxide high temperature anneal, as shown in Fig. 4.

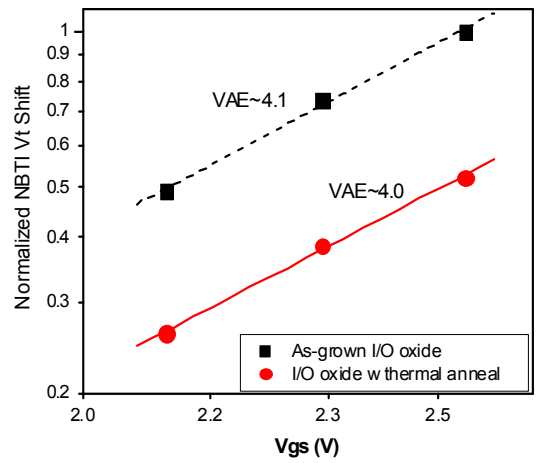

Fig. 1. The Voltage Acceleration Exponent (VAE) of DC NBTI Vth shift on I/O pMOSFETs with and without additional gate oxide thermal anneal

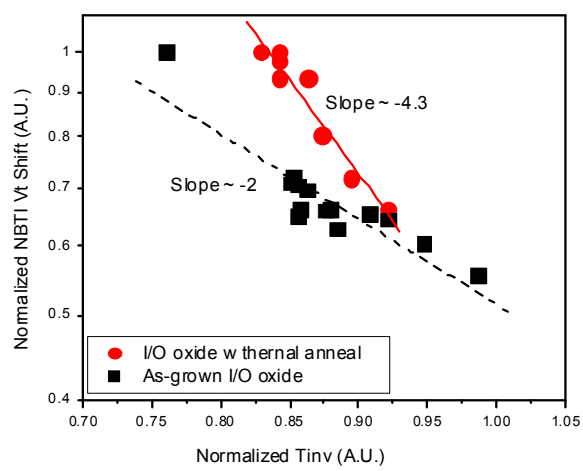

Fig. 2 DC NBTI V_{th} shift shows different dependence on T_{inv} thickness for I/O pMOSFETs with and without additional gate oxide thermal anneal at the same Vg at stress

B. Comparison of VAE, Tinv dependence, and recovery of NBTI under AC bias stress

To further validate this hypothesis, we checked the NBTI VAE and T_{inv} dependence under AC bias. The AC NBTI V_{th} shift showed consistent VAE and T_{inv} dependence with DC results, for both processes respectively, as given in Figs. 5 and 6.

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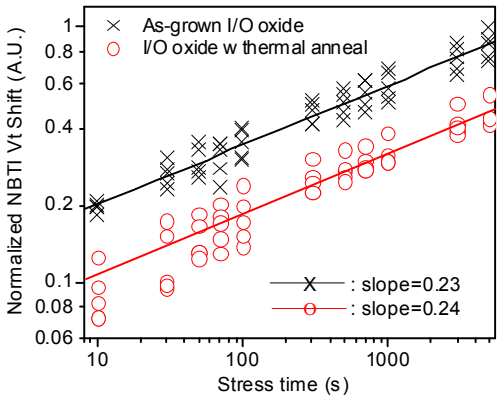

Fig. 3. DC NBTI V_{th} shift is reduced by ~2X on devices with extra thermal anneal gate oxide process for the same T_{inv} thickness.

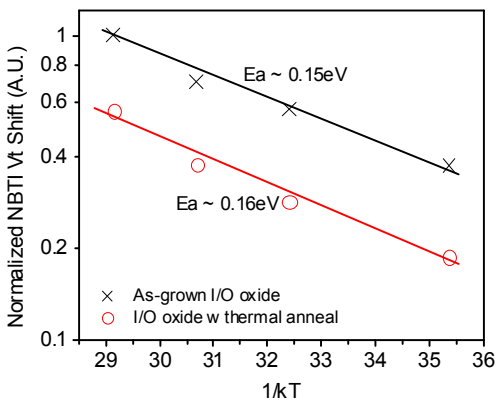

Fig. 4. Similar DC NBTI activation energy observed on devices with and without additional gate oxide thermal anneal.

The similar T_{inv} dependence observed in both processes under DC and AC NBTI conditions further indicates that this defect component responsible for the NBTI aging on device with as-grown gate oxide, is not associated to a recoverable fast hole traps (with similar capture and emission time) activation typically reported in DC conditions. Since this fast hole trap activation does not contribute to AC NBTI V_{th} shift, a $\Delta V_{th} \sim T_{inv}^{-4}$ would have been expected. These results further suggest that the damaging defects causing high NBTI V_{th} shift in the as-grown gate oxide device are most likely process induced deep hole traps with very short capture time (<< 1 s) and very long emission time to contribute as a quasi permanent part of NBTI aging. Our results support the following NBTI model in both I/O oxide processes with and without the additional high temperature anneal, for both DC and AC stress conditions.

$$\Delta V_{th} = A \times \left(\frac{V_g}{T_{inv}}\right)^4 \times e^{-0.15/kT} \times t^{0.24}$$

We explain the T_{inv}^{-2} NBTI dependence of the as-grown thermal gate oxide process by assuming the quasi permanent component has a concentration density or energy distribution such that the prefactor A is dependent on T_{inv} as T_{inv}^2. The consequent DC NBTI enhancement yields an apparent violation of the NBTI dependence on E_{ox}. The high temperature thermal anneal greatly passivates these defects present in the as-grown gate oxide, such that NBTI is improved significantly and follows the normal E_{ox} dependence (the prefactor parameter A is not T_{inv} dependent), as given in Fig. 8. Furthermore, we examined the NBTI recovery behavior closely for both DC and AC bias stress conditions. The similar NBTI recovery behavior after DC stress and the minimal recovery observed after AC stress (see Fig. 9) for both processes indicates that the NBTI physical mechanisms are basically the same for the devices with both processes, and the defects (deep hole traps and/ or interface states) part of the quasi permanent component contribute to the NBTI V_{th} shift in the as-grown oxide device, which are responsible for the weak T_{inv} dependence.

The comparison between the DC and AC time evolution, as shown in Figs. 3 and 7, suggests that this 2x reduction is explained by reduce contribution of the quasi permanent part with the addition gate oxide thermal anneal.

The observation that a similar damage under Positive Bias Temperature Instability (PBTI) stressing observed in I/O nMOSFETs with both gate oxide processes suggests that the reported NBTI Tinv dependence is peculiar of the NBTI induced damage (quasi permanent part) at the IL/Silicon interface. As shown in Fig. 10, there is no difference in PBTI Vt shift comparing the two gate oxide processes, since PBTI is dominated by electron trapping in the HK oxide. The PBTI equivalency is explained by the observation that the electron traps generated during the PBTI stress in HK oxide are not passivated by the extra anneal.

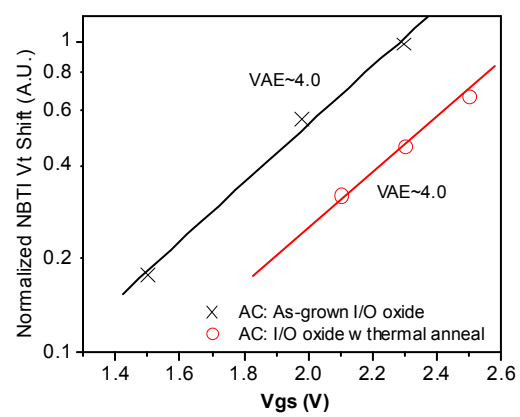

Fig. 5. The Voltage Acceleration Exponent (VAE) of AC NBTI Vth shift on I/O pMOSFETs with and without additional gate oxide thermal anneal

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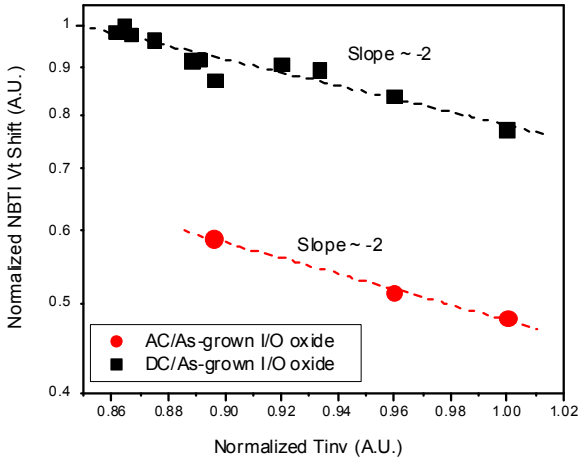

Fig. 6. Same weak T_{inv} dependence observed under AC and DC NBTI for devices without additional gate oxide thermal anneal.

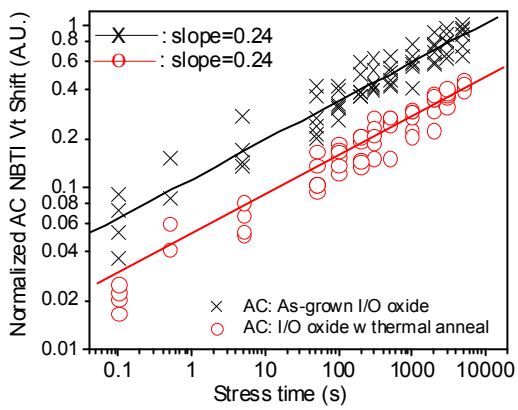

Fig. 7 AC BTI V_{th} shift is also reduced by ~2X on devices with gate oxide with extra thermal anneal processing for the same T_{inv} thickness.

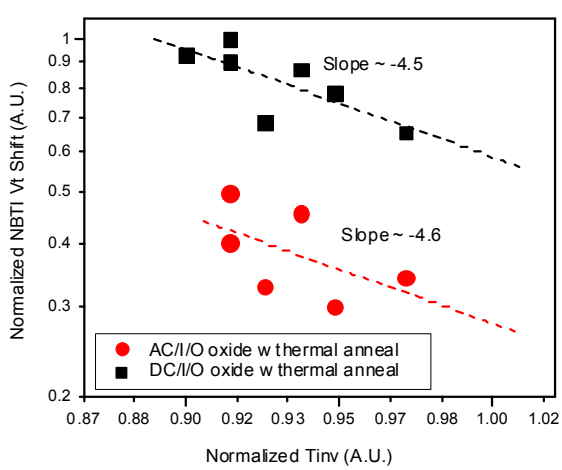

Fig. 8 Normal T_{inv} dependence observed under AC and DC NBTI for devices with additional gate oxide thermal anneal.

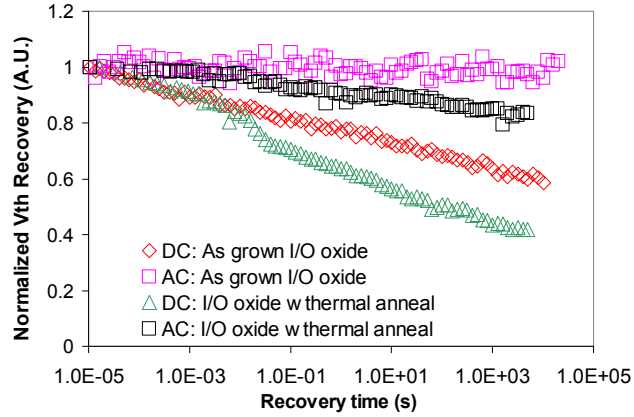

Fig. 9. NBTI V_{th} shift recovery behavior comparison under AC and DC bias for devices with and without additional gate oxide thermal anneal.

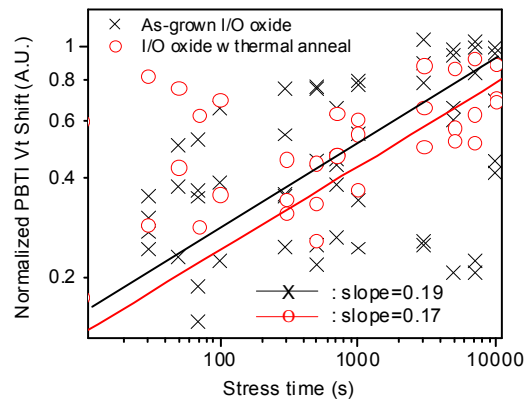

Fig. 10. Equivalent DC PBTI induced Vt shift is measured for nMOSFET devices with and without additional gate oxide thermal anneal.

C. *Implication on technology qualification methodology*

As discussed above, due to the abnormal weak T_{inv} dependence of NBTI, if one assumes the same E_{ox} and VAE dependence of NBTI for the as-grown thermal gate oxide process, which is a common methodology for a technology qualification, apparently the NBTI V_{th} shift would be underestimated to result in an over-optimistic projection. Therefore, to model a new process or qualify a new technology process correctly, one needs to evaluate both the gate voltage V_{gs} dependence for a fixed T_{inv} and the gate oxide field (E_{ox}~V_{gs}/T_{inv}) dependence, to ensure they follow the same value.

IV. Conclusion

The as-grown thermal I/O oxide gives an abnormally weak T_{inv} dependence of both DC and AC NBTI. Following a high temperature thermal anneal treatment, the NBTI degradation is reduced significantly and the V_{th} shift recovers the E_{ox} expected dependence. It is demonstrated that the defects causing the

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weak NBTI T_{inv} dependence in the as-grown oxide attribute to a permanent component and are not related to fast hole trap activation in the oxide. This permanent component contribution increases with T_{inv}, and yields an apparent violation of the NBTI dependence on E_{ox}. Further evidence is given on two independent and decoupled (fast hole trap activation and permanent part) contributors to the NBTI damage. The implication of these findings on a technology qualification methodology is that one needs to evaluate independently both the V_g and T_{inv} dependence of NBTI V_{th} shift for a given process to confirm the expected E_{ox} dependence. An immature process, in fact, may not yield the normal E_{ox} dependence.

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